

Overview

Work for a Quantitative research club at UBC

Includes building an earthquake prediction model

Main ideas

Models in order of their prediction capabilities (worst to best)

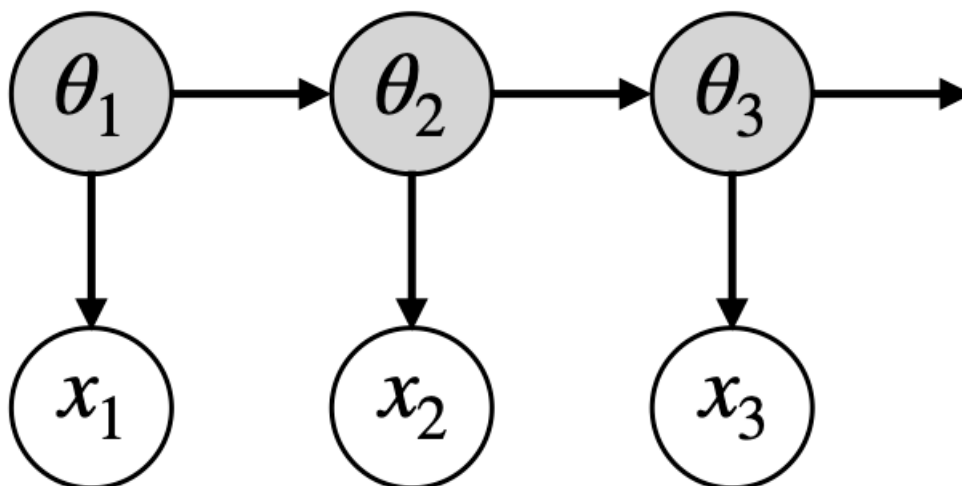
1. Classic ML: Random Forest or XGBoost forest for prediction using seismic data
2. *HMM*: Gaussian and Poisson univariate and multivariate *HMM* works
3. *GEV*: modelling severe earthquakes with high magnitudes according to extreme tailed distributions
4. *PSHA*: Probabilistic Seismic Hazard Assessment model

All of these models were fine-tuned and used for earthquake and seismic prediction which would then be used for a portfolio optimisation model as a factor. Improvement on the the optimisation model will be backtested and reported in the paper.

I helped with a different models but mainly focused on *HMM* and the notes for it is reported below

Hidden Markov Models for Earthquake Prediction: GaussianHMM Analysis

Hidden Markov Model



Overall summary

Main Idea:

- split data into training (80%) and testing (20%) chronologically
- train Gaussian *HMM* based on the training data
- use best performing *HMM* parameter (based on *BIC*) to forecast values
 - the best X_{train} is based on the time series cross validation with `n_splits=3`
 - otherwise it was always choosing the highest number of states and fitting to it
- Get the $\mathcal{L}$, *AIC*, *BIC* of the best performing model on the testing data

$\mathcal{L}$: -3835.579679558612
 AIC : 8043.159359117224
 BIC : 9162.3785555473

- Use the *HMM* to fit states on the testing data based on feature thresholds and differing criteria
 - criteria that fits the states on the test features the best is chosen based on most accuracy
- Compare fitted value against forecasted values to get the following:

Confusion Matrix:

```

[[ 0  0  0  0  0  0  0  0  0  0  0]
 [ 9 21  1  0  0 153  1  0 159  0 29]
 [ 5  0  3  0  0  90  0 31  14 98  0]
 [40  0  0 122  0 366 17  0 171  0 489]
 [18  0  0  0 35  0  0  0  0  0  0]
 [68  0  0  0  0 113  0 11  51  2  0]
 [ 4  1  0  3  0  40 233  0  42  0 223]
 [ 3  0  0  0  0  0  0 21  0  0  0]
 [21  0  0  0  0  0  0  2  53  0  0]
 [ 5  0  0  0 16  0  0  5  0 100  0]
 [ 9  0  0  0  0  0  0  0  20  0 115]]
  
```

Weighted Average Classification Report:

{'precision': 0.7879553180628392, 'recall': 0.26904055390702275, 'f1-score': 0.2675312452338454}

For reference, the training metrics were:

$\mathcal{L}$: -5590.1346817662425
 AIC : 11552.269363532485
 BIC : 12679.881862766959

Tried with univariate models on just the counts for Poisson (and also Gaussian):

Poisson:

$\mathcal{L}$: -2536.0145200794263
 AIC : 5434.029040158853
 BIC : 6403.977432758818

Gaussian:

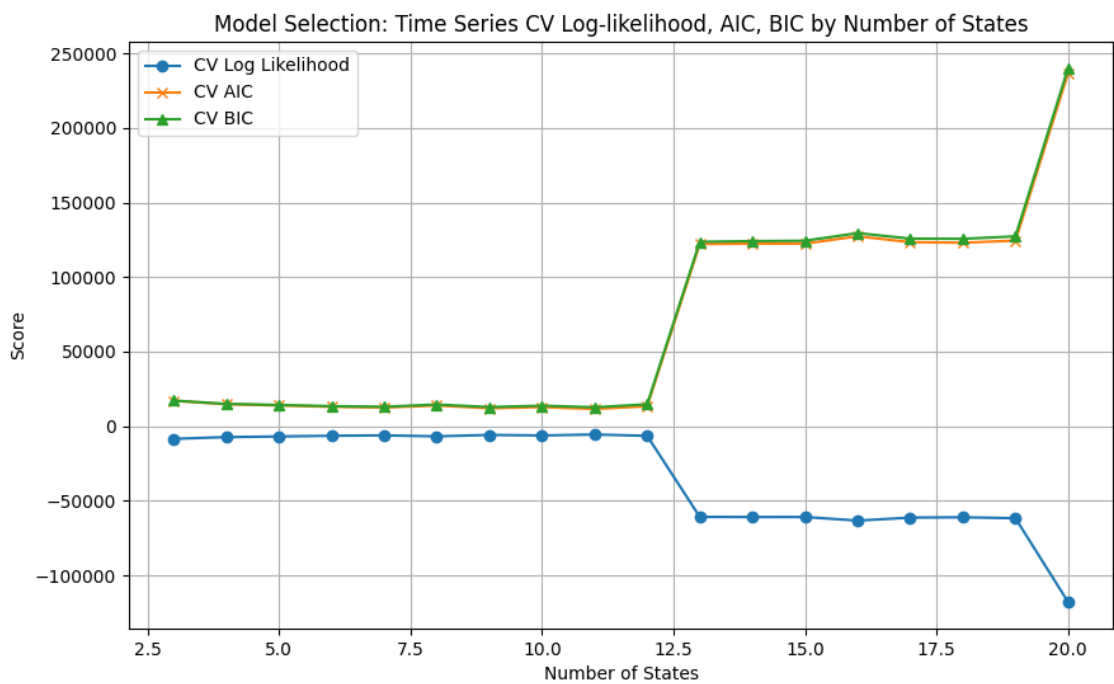
$\mathcal{L}$: 5456.240959411002
 AIC : -10230.481918822004
 BIC : -8403.120582487261

HOWEVER, there are couple of issues with just using univariate data based on counts of above a threshold

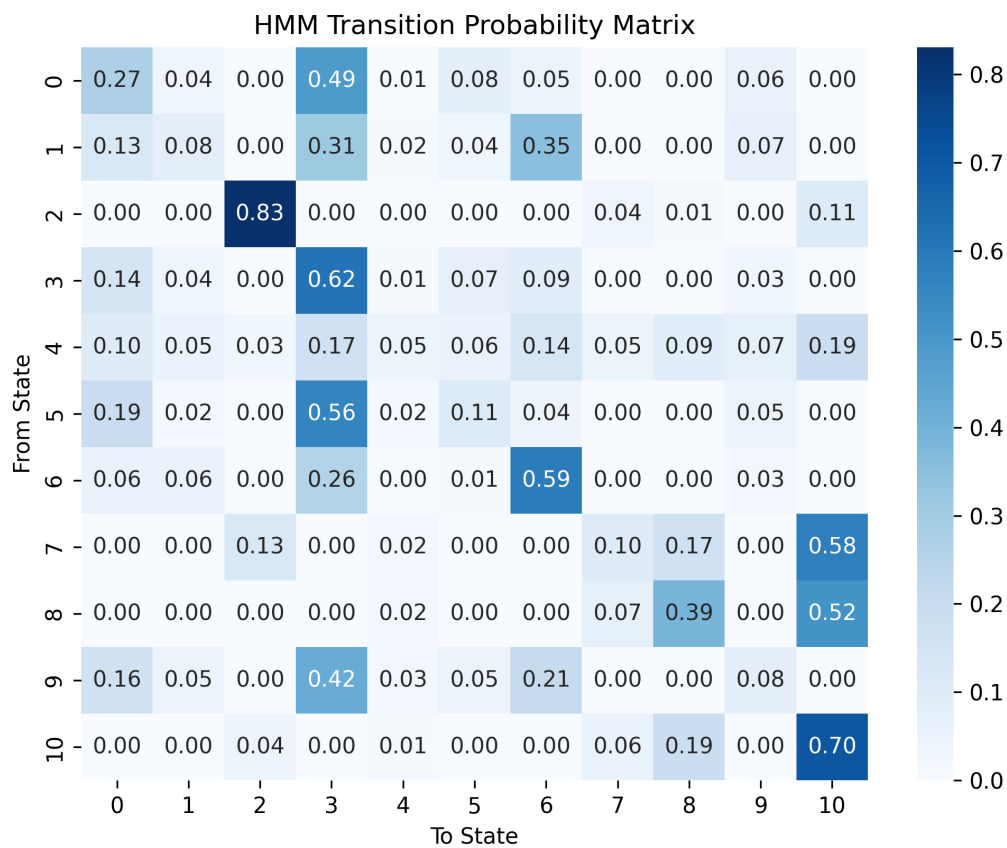
1. Loss of risk-critical information: A univariate count model ignores magnitude, depth, location, and energy release patterns that are essential for assessing actual financial/infrastructure risk
 - i. a single magnitude 8.0 earthquake has vastly different hazard than three magnitude 6.0 events, despite both scenarios having different "counts"
2. No actionable insights for risk mitigation: Risk modelling or portfolio optimisation models will need to understand what types of seismic conditions drive risk (shallow vs deep events, proximity to assets, energy buildup patterns) to make informed hedging or allocation decisions - a simple count prediction provides no guidance on which geographic regions, asset types, or time horizons require different risk premiums

Model interpretation

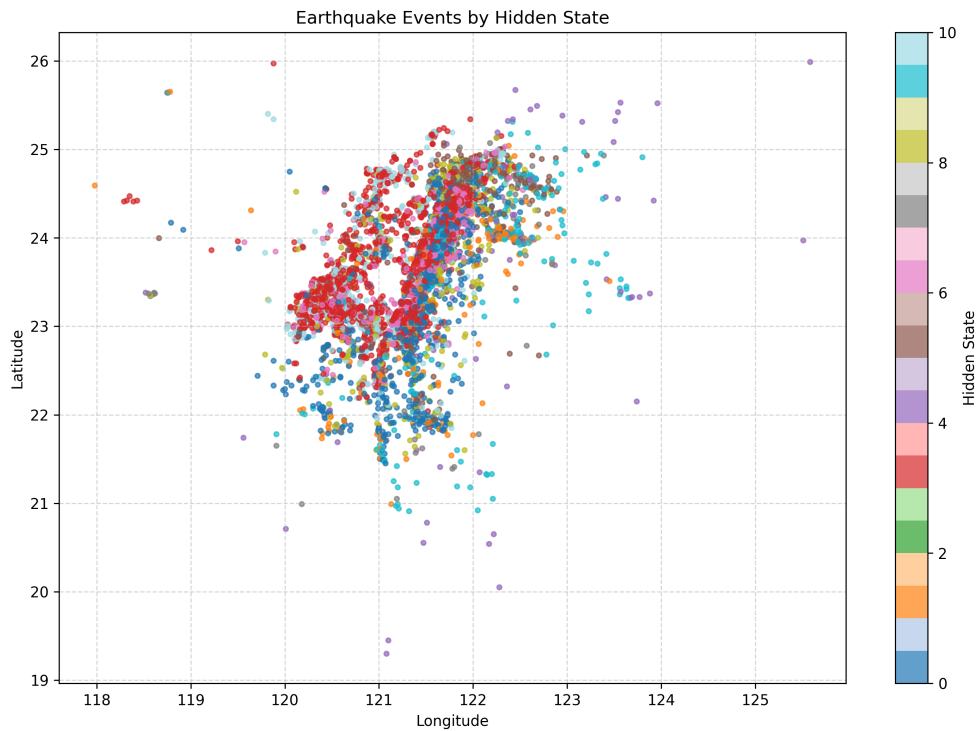
With the plots, we can see that after state 11, the states were not converging:



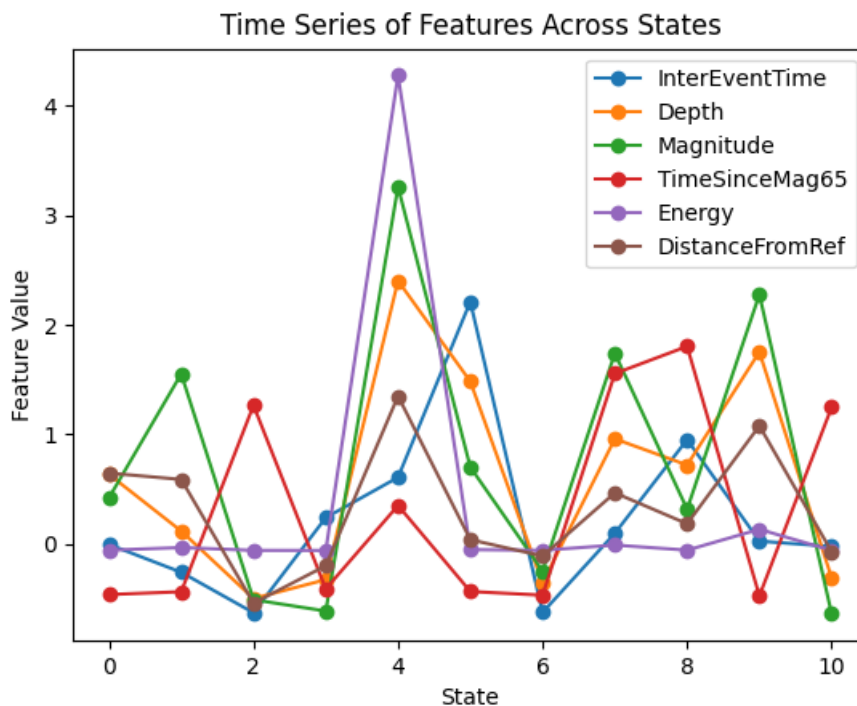
We can see that the matrix is quite sparse:



The model becomes a bit hard to interpret when looking geographically:



So, looking at the features we get:



Analysis

Transition matrix

- Diagonal dominance in transition matrix: Strong self-transition probabilities suggest states tend to persist, indicating earthquake activity has temporal clustering or "stickiness"
 - the aftershocks
- Limited State Connectivity: Most states only transition to 2-3 other states, suggesting distinct seismic regimes with restricted pathways between them
- Potential Absorbing States: Some states may act as temporary "attractors" in the seismic cycle?

- Like state 10

Geographic interpretation

- There is a lot of spatial mixing,
- could be that states are more clustered due to temporal or magnitude based events rather than graphically (not clustered near fault line)
 - maybe state 6 could be related with severe earthquake
- more symptomatic of the univariate distribution with counts and no other features
- Different earthquake "phases" or "regimes" can occur across the entire study region

Feature state relation

The following is my interpretation from reading about how earthquakes work and trying to interpret the graph, use it as reference with caution ⚠

States 3-6: Moderate Severe Earthquake Phase

- **Magnitude:** Elevated magnitude levels (4.5-5.5 range)
- **Inter-event Time:** Moderate to high inter-event times, suggesting these occur during building stress phases
- **Depth:** Mixed depth distribution, indicating both shallow and deeper moderate events
- **Time Since Mag 6.5:** Moderate values, representing intermediate phases in seismic cycles
- **Energy:** Elevated energy release corresponding to moderate-severe events
- **Distance from Reference:** Distributed across various distances from reference point

Seismic Interpretation: These states capture the transitional phase between background seismicity and major events - likely representing foreshock sequences, moderate mainshocks, or stress redistribution following major earthquakes.

States 8-10: Major Earthquake/Immediate Aftermath Phase

- **Magnitude:** Highest magnitude values (5.5+ range), representing major seismic events
- **Inter-event Time:** Very low values, indicating rapid succession of events (classic aftershock pattern)
- **Depth:** Relatively shallow depths, making these events particularly hazardous
- **Time Since Mag 6.5/5:** Very low values, confirming these states occur immediately following or during major earthquake sequences
- **Energy:** Peak energy release values, representing maximum seismic energy discharge
- **Distance from Reference:** Concentrated patterns suggesting spatial clustering of major events

Seismic Interpretation: These states represent the most critical seismic phases - major mainshocks (State 10) and their immediate aftershock sequences (States 8-9), characterized by high magnitude, shallow depth, and temporal clustering.

The *HMM* pipeline is given in much more detail below:

1. Gaussian Hidden Markov Models (GaussianHMM)

1.1 Mathematical Foundation

A Hidden Markov Model (HMM) is a statistical model that assumes the system being modeled is a Markov process with unobserved (hidden) states. For earthquake prediction, we use GaussianHMM where the observable features follow a multivariate Gaussian distribution.

Model Components

A GaussianHMM is characterized by:

- **Number of hidden states:** N
- **Initial state probabilities:** $\pi = \{\pi_i\}$ where $\pi_i = P(q_1 = S_i)$
- **State transition probabilities:** $A = \{a_{ij}\}$ where $a_{ij} = P(q_{t+1} = S_j | q_t = S_i)$
- **Emission probabilities:** Gaussian distributions for each state

Severe Event Definition:

$$\text{Severe Event} = \begin{cases} 1 & \text{if } (M \geq 6 \text{ and } D \leq 100) \text{ or } (M \geq 5.5 \text{ and } D \leq 10) \\ 0 & \text{otherwise} \end{cases}$$

Where M is magnitude and D is depth in kilometers.

Mathematical Formulation

For a sequence of observations $O = \{o_1, o_2, \dots, o_T\}$ and hidden states $Q = \{q_1, q_2, \dots, q_T\}$:

State Transition Probability:

$$P(q_{t+1} = j | q_t = i) = a_{ij}$$

Emission Probability (Gaussian): For state i , the observation o_t follows a multivariate Gaussian distribution:

$$b_i(o_t) = P(o_t | q_t = i) = \mathcal{N}(o_t; \mu_i, \Sigma_i)$$

Where:

$$\mathcal{N}(o_t; \mu_i, \Sigma_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left(-\frac{1}{2} (o_t - \mu_i)^T \Sigma_i^{-1} (o_t - \mu_i) \right)$$

- μ_i : mean vector for state i
- Σ_i : covariance matrix for state i
- d : dimensionality of observations

Joint Probability

The joint probability of observations and hidden states:

$$P(O, Q | \lambda) = \pi_{q_1} \prod_{t=1}^{T-1} a_{q_t, q_{t+1}} \prod_{t=1}^T b_{q_t}(o_t)$$

Where $\lambda = \{\pi, A, B\}$ represents the model parameters.

Forward-Backward Algorithm

Forward variable:

$$\alpha_t(i) = P(o_1, o_2, \dots, o_t, q_t = i | \lambda)$$

Recursive computation:

$$\alpha_{t+1}(j) = \left[\sum_{i=1}^N \alpha_t(i) a_{ij} \right] b_j(o_{t+1})$$

Backward variable:

$$\beta_t(i) = P(o_{t+1}, o_{t+2}, \dots, o_T | q_t = i, \lambda)$$

Likelihood computation:

$$P(O | \lambda) = \sum_{i=1}^N \alpha_T(i)$$

1.2 Feature Engineering for Earthquake Data

Our GaussianHMM uses the following earthquake features:

1. **Inter-event Time:** Time between consecutive earthquakes (days)
2. **Depth:** Earthquake depth (km)
3. **Magnitude:** Earthquake magnitude (Richter scale)
4. **Time Since Magnitude 6.5+:** Days since last major earthquake
5. **Energy:** Seismic energy release: $E = 10^{1.5M+4.8}$ Joules
6. **Distance from Reference:** Haversine distance from reference point (24.785°N, 121.006°E)

Energy Budget Computation

Energy budget analysis using rolling window approach:

$$\text{Energy Budget Ratio} = \frac{\sum_{i=t-w}^t E_i}{\bar{E} \times w}$$

Where:

- E_i : Energy of earthquake i
- $\bar{E}$: Average daily energy release
- w : Window size (30 days)

2. Validation and Testing Methodology

2.1 Model evaluation metrics

Log-Likelihood:

$$\mathcal{L}(\lambda) = \log P(O|\lambda) = \log \sum_Q P(O, Q|\lambda)$$

Akaike Information Criterion (AIC):

$$\text{AIC} = 2k - 2\mathcal{L}(\lambda)$$

Bayesian Information Criterion (BIC):

$$\text{BIC} = k \log(n) - 2\mathcal{L}(\lambda)$$

Where:

- k : Number of model parameters
- n : Number of observations
- For GaussianHMM: $k = N(N - 1) + (N - 1) + Nd + Nd = N^2 + 2Nd - 1$

2.2 Fitting states to test O

States are categorized using statistical bounds:

$$\text{Lower Threshold}_i = \mu_i - 2\sigma_i$$

$$\text{Upper Threshold}_i = \mu_i + 2\sigma_i$$

Categorization Criterion: An observation is assigned to state j if at least $\eta \times 100\%$ of its features fall within the thresholds:

$$\text{Assigned State} = \arg \max_j \left\{ \frac{\sum_{d=1}^D \mathbb{I}[\mu_{j,d} - 2\sigma_{j,d} \leq x_d \leq \mu_{j,d} + 2\sigma_{j,d}]}{D} \geq \eta \right\}$$

Where $\mathbb{I}[\cdot]$ is the indicator function and $\eta \in [0.5, 1.0]$ is the criteria threshold.

2.2.1 Classification Performance Metrics

Accuracy: Percentage of correctly classified states **Precision, Recall, F1-score:** Standard classification metrics **Confusion Matrix:** Cross-tabulation of predicted vs actual states

fitted states were compared against forecasted states for getting an estimate of state prediction accuracy